Practice Project

FitZone Midwest: Sports Retail Sales Analysis

A Power BI skills-building exercise to prepare for Capstone 3

Theme: Sporting goods retail | 2-year dataset (2024–2025) | Midwest Region | 14 stores + online

Files included with this practice project

FitZone_Practice_Sales_Data.xlsx	Main dataset — 6 sheets: In-Store Transactions, Online Orders, Store Locations, Products, Regional Management, Membership Tiers
FitZone_Product_Reference.txt	Pipe-delimited product reference — 49 products with brand, key feature, top endorser, and target audience. Import and join on Product ID.
This document (.docx)	Full project brief, guidance, skills checklist, data dictionary, reflection questions, and expected outcomes.

Section 1 — Scenario & Project Overview

About FitZone

FitZone is a regional sporting goods and wellness retailer operating 14 physical stores across Illinois, Indiana, and Ohio, with an e-commerce presence shipping to 10 Midwest states. Founded in 2020, FitZone targets serious fitness enthusiasts and casual gym-goers alike, offering products across seven categories: Sporting Goods, Apparel & Gear, Nutrition & Supplements, Electronics & Tech, Books & Media, Recovery & Wellness, and Footwear.

Your Role

You are a data analyst supporting Patricia Holloway, the Midwest Regional Director. Patricia has asked you to build a Power BI report analyzing FitZone's sales performance over the past two years (2024–2025). She wants to share this report with state managers and store managers at the upcoming Midwest Regional Leadership Retreat.

Your Deliverable

A clean, professional Power BI report with at least two pages, containing the four required visuals, clear page headers, and at least one interactive slicer. The report should tell a clear story about FitZone's business performance and help Patricia's team make better decisions.

Section 2 — Project Requirements

Required visuals (all four must appear in your report)

#	Visual type	What it should show

1	**Line chart**	Monthly sales trend for the full 2-year period (Jan 2024 – Dec 2025), combining in-store and online channels
2	**Column or bar chart**	Total sales by product category across the Midwest region
3	**Pie or donut chart**	Relative sales contribution of each state in the region (Illinois, Indiana, Ohio + online states)
4	**Table / matrix**	Top-selling products for the region (general fitness products only, excluding textbooks/media), including product name, brand, and endorser from the reference file

Additional report requirements

- Report must have at least 2 pages, each with a descriptive text header
- Include at least one slicer to enable interactive filtering
- All axis labels and chart titles must be readable and meaningful (no default 'Sum of…' labels)
- Format currency values consistently throughout the report
- Save your Power BI file as: YourLastName_FitZone_Practice.pbix

Section 3 — Data Files & Data Dictionary

File 1: FitZone_Practice_Sales_Data.xlsx (6 sheets)

Sheet name	Sheet role / key columns
In-Store Transactions	Primary fact table — date (serial), Store ID, Product ID, Category, Unit Price, Units Sold, Sale Amount formula
Online Orders	Second fact table — Year, Month, Day split, Product ID, Order Total, Ship-to State, Shipping Method
Store Locations	Store dimension — Store ID, City, State, Region, Store Manager, Open Date
Products	Product dimension — Product ID, Name, Category, Subcategory, Brand, MSRP, Is Featured flag
Regional Management	Region/state hierarchy — Region, Director, State, State Manager, Territory Notes
Membership Tiers	Reference table — Tier, minimum spend, discount %, shipping benefits (bonus analysis context)

File 2: FitZone_Product_Reference.txt (pipe-delimited)

This file is a plain-text reference file that must be imported into Power Query and joined to your sales data using the Product ID column. It contains 49 products with 7 fields per row:

Product ID	Matches the Product ID column in both sales tables — use this as your join key
Product Name	Full product name
Category	Matches the Category in the Products sheet
Brand	Manufacturer/brand name (e.g. ProFlex, VitalTrack)

Key Feature	One-line product description — useful for the top-sellers table
Top Endorser	Athlete or influencer who endorses this product — include in your top-sellers visual
Target Audience	Primary customer segment

Import tip: In Power Query, use Get Data → Text/CSV. Change the delimiter from Comma to Custom and enter | (pipe). Skip the first 4 header rows by using 'Skip top rows' under Advanced Options, or promote headers after import. The file is pipe-delimited, not comma-delimited.

Section 4 — Data Quirks to Find & Fix

The dataset contains intentional data quality issues that mirror what you will encounter in Capstone 3. Finding and resolving these is part of the exercise. Use the table below as a guide.

Issue	How to fix it in Power Query
Date stored as Excel serial number	In-Store Transactions: the Transaction Date column is a numeric serial (e.g. 45292 = Jan 1, 2024). Change the column type to Date in Power Query — Power BI will interpret the serial correctly.
Date split into Year/Month/Day columns	Online Orders: three separate columns instead of a single date. Use Add Column → Custom Column with Date.From(#date([Year],[Month],[Day])) to create one unified date field.
Blank / null Category values	A small number of In-Store rows have a null or empty Category. Use Replace Values or Remove Rows in Power Query. Decide whether to drop them or label them 'Uncategorized' — justify your choice.
Negative Unit Price (data entry error)	A few rows have a negative Unit Price. Filter or replace these — a price below $0 is invalid. Check whether the Sale Amount formula also produces a negative and clean accordingly.
Store ID entered as 'N/A' text	A handful of In-Store rows have the text 'N/A' instead of a numeric Store ID. These cannot be joined to Store Locations. Either remove these rows or flag them as 'Unknown' in a separate step.
TXT file requires custom delimiter	The product reference file uses | as its delimiter, not a comma. You must specify this when importing — Power Query will default to comma. Also handle the 4-line header before the data starts.

Section 5 — Step-by-Step Guidance

Step 1 — Explore the data before you clean anything

Before touching Power Query, spend 10–15 minutes reviewing both files. Open the Excel file and scroll through each sheet. Open the TXT file in Notepad or VS Code and read the header rows. Ask yourself:

- What columns exist in each sheet? Do any look unusual?
- What column will I use to join the Products sheet to the sales data?
- How many unique categories, states, and stores are there?
- Are there obvious formatting issues I can see just by scrolling?

Sketch a rough diagram of how the tables relate to each other before you start. This saves time later when building relationships in Model View.

Step 2 — Connect data sources and begin Power Query

- Get Data → Excel Workbook → select FitZone_Practice_Sales_Data.xlsx → load all 6 sheets
- Get Data → Text/CSV → select FitZone_Product_Reference.txt → set delimiter to | → skip first 4 rows → promote headers
- Fix the date column in In-Store Transactions (change type to Date)
- Create a merged Date column in Online Orders from Year + Month + Day
- Resolve the data quirks listed in Section 4 — do this before creating any visuals
- Merge (join) the Products query into the sales queries on Product ID to add category labels
- Merge the Product Reference TXT into your sales data on Product ID to add Brand and Endorser fields

Save frequently. Each time you find a new issue, note what it was and how you fixed it — you will need this for your reflection.

Step 3 — Build the data model

- Create a Date/Calendar table spanning Jan 1 2024 to Dec 31 2025. Mark it as a date table.
- In Model View, create relationships: Date table → both fact tables; Products → both fact tables on Product ID; Store Locations → In-Store Transactions on Store ID
- Write your key DAX measures: Total Sales = SUM() across both fact tables; Sales by Channel; YTD Sales
- Consider a RANKX measure to rank top-selling products by revenue

Check your model: can you drop a Category onto a visual and see the correct sales broken out? If not, a relationship is wrong or missing.

Step 4 — Build the report visuals

- Page 1 suggested: Overview — line chart (trend), pie/donut (by state), header, Year slicer
- Page 2 suggested: Product Analysis — bar chart (by category), top-sellers table, Category slicer
- Format every chart: rename axes, add a chart title, remove default 'Sum of' labels
- Add at least one slicer. Test that it filters all visuals on the page correctly.
- Use consistent colors that feel different from the Capstone 3 report — this is practice, make it your own

Section 6 — Skills Checklist

Use this checklist to track your progress. All items marked [Core] map directly to a Capstone 3 requirement. Items marked [Stretch] go beyond the minimum and will strengthen your skills further.

Data cleaning & Power Query

- ☐ **[Core] Import an Excel workbook with multiple sheets into Power Query**
  Use Get Data → Excel Workbook. Load each of the 6 sheets as its own query. Rename queries clearly.
  Practice tip: Try loading the 6 sheets and naming them: StoreSales, OnlineOrders, StoreLocations, Products, RegionalMgmt, MemberTiers

- ☐ **[Core] Import a delimited text file with a non-standard delimiter**
  Use Get Data → Text/CSV. Change the delimiter from Comma to Custom and type |. Handle the 4-line header block.
  Practice tip: After import, check that 7 columns appear with the correct headers: Product ID, Product Name, Category, Brand, Key Feature, Top Endorser, Target Audience

- [ ] **[Core] Fix a date stored as an Excel serial number**
  Change the column type to Date in Power Query. Power BI correctly interprets Excel date serials when the type is set.
  Practice tip: After changing the type, confirm that the earliest date shows as 1/1/2024 and the latest as 12/31/2025

- [ ] **[Core] Merge three separate date columns into one**
  In Online Orders, use Add Column → Custom Column: Date.From(#date([Year],[Month],[Day])). Remove the original three columns.
  Practice tip: Spot-check: a row with Year=2024, Month=6, Day=15 should produce a Date of 6/15/2024

- [ ] **[Core] Handle null/blank values and data entry errors**
  Fix blank Category rows, negative Unit Price rows, and text 'N/A' Store ID rows. Use Replace Values, Filter Rows, or conditional columns.
  Practice tip: After cleaning, your Category column should have zero nulls and zero #N/A values. Unit Price should have no negatives.

- [ ] **[Core] Merge queries to enrich data (JOIN equivalent)**
  Use Merge Queries to join Products into sales tables on Product ID. Then join the TXT reference file to get Brand and Endorser.
  Practice tip: After merging, your sales rows should include Category, Brand, and Top Endorser alongside the transaction data

- [ ] **[Stretch] Append two fact tables into a single combined sales table**
  Use Append Queries to stack In-Store Transactions and Online Orders after normalizing their column names and date formats.
  Practice tip: A combined table makes it easier to write a single Total Sales measure and build a unified date axis

Data modeling

- [ ] **[Core] Create and mark a Date/Calendar table**
  In DAX: DateTable = CALENDAR(DATE(2024,1,1), DATE(2025,12,31)). Add Year, Month, Month Name columns. Mark as date table in Model View.
  Practice tip: Verify the table by dropping Month Name onto a line chart — months should appear in calendar order, not alphabetical

- [ ] **[Core] Build a star-schema model with correct relationships**
  In Model View: Date → both fact tables; Products → both fact tables; StoreLocations → StoreSales. Set filter direction to Single for all.
  Practice tip: Cross-filter test: click a state in your pie chart — does the line chart update to show only that state's trend?

- [ ] **[Core] Write a Total Sales DAX measure**
  Total Sales = SUM(StoreSales[Sale Amount]) + SUM(OnlineOrders[Order Total]). Use this measure everywhere instead of implicit aggregation.
  Practice tip: Confirm the measure gives the same result as dragging the raw Sale Amount column onto a visual — then delete the implicit measure

- [ ] **[Stretch] Write a Sales by Channel measure**
  Use CALCULATE() with a filter to split Total Sales into In-Store vs. Online. Useful for a stacked column chart.
  Practice tip: In Store Sales = CALCULATE([Total Sales], FILTER(StoreSales, StoreSales[Store ID] <> "N/A"))

- [ ] **[Stretch] Use RANKX to rank top-selling products**
  TopProductRank = RANKX(ALL(Products[Product Name]), [Total Sales]). Use this in a visual-level filter to show only the Top 10.
  Practice tip: Add a visual-level filter: TopProductRank is less than or equal to 10

Visuals & report design

- [ ] **[Core] Build a line chart with a date hierarchy**
  X-axis: Date hierarchy from your calendar table (Year → Quarter → Month). Y-axis: Total Sales measure. Enable drill-down to reach monthly detail.
  Practice tip: Can you click the drill-down arrow and reach individual months? Does the line show a reasonable trend across 2 years?

- [ ] **[Core] Build a column/bar chart with category dimension**
  Category on the axis, Total Sales on the value. Sort descending by Total Sales. Check that #N/A categories are gone after cleaning.
  Practice tip: Expected top categories: Footwear and Electronics & Tech typically drive the highest revenue in practice data

- [ ] **[Core] Build a pie or donut chart by state**
  State dimension on the legend, Total Sales on values. Include both in-store states (IL, IN, OH) and online ship-to states.
  Practice tip: Illinois should have the largest slice (most stores). Online-only states like Michigan and Minnesota should also appear.

- [ ] **[Core] Build a top-sellers table with endorser names**
  Columns: Product Name, Brand, Top Endorser, Total Sales. Filter to exclude Books & Media category. Sort by Total Sales descending.
  Practice tip: Sierra Watts (powerlifter) and Tanya Osei (marathon runner) products should appear near the top based on the dataset

- [ ] **[Core] Add descriptive page headers to every page**
  Insert a Text Box or formatted shape at the top of each page. Use a clear title. At minimum: Page 1 — Sales Overview, Page 2 — Product Insights.
  Practice tip: Ask: if a store manager opened this report cold, would they immediately know what page they are on?

- [ ] **[Core] Add and test a slicer**
  Add a Year slicer (2024 vs 2025) or a State slicer. Confirm that selecting a value updates all visuals on the page.
  Practice tip: Test edge case: what happens if you select both years? Does the data double-count? It should not.

- [ ] **[Core] Rename axes and titles (no default 'Sum of' labels)**
  In the Format pane, update every chart's title, X-axis label, and Y-axis label. Currency values should include a ($) hint.
  Practice tip: Before: 'Sum of Sale Amount'. After: 'Total Sales ($)'. Check every visual before you consider it done.

Section 7 — Reflection Questions

After completing your report, write brief answers to the questions below. These mirror the reflection you will give in your Capstone 3 video presentation. There are no right or wrong answers — the goal is to practice articulating your analytical decisions.

Part A — Data cleaning decisions

1. What was the most unexpected data quality issue you found? How did you discover it, and what did you do to fix it?
2. When you found rows with blank Category values, what did you decide to do with them — remove or relabel? Why? How might your choice affect the analysis?

3. When combining In-Store and Online Sales, what challenges did you face in making the two tables compatible? What column names did you have to normalize?
4. How did importing the pipe-delimited TXT file differ from importing the Excel file? What extra steps were required?

Part B — Modeling decisions

5. Draw or describe your final data model. How many tables does it have? What are the relationships between them?
6. Why is it better to write an explicit DAX measure (Total Sales = SUM(...)) rather than just dragging a column onto a visual? What problems can implicit measures cause?
7. What does 'marking a table as a date table' do in Power BI, and why does it matter for your line chart?

Part C — Insights from the data

8. Which product category generated the highest total revenue over the 2-year period? Does this surprise you? What could explain it?
9. Looking at the sales trend line chart, do you see any seasonal patterns? When are sales highest? When are they lowest? What business factors might explain this?
10. Which state contributed the most to in-store revenue? Which online ship-to state had the most orders? Are these the same state?
11. If you were presenting to Patricia Holloway (the Regional Director), what is the single most important insight you would highlight, and why?
12. What question does the data raise that you cannot answer from these datasets alone? What additional data would help?

Section 8 — Expected Outcomes & Reference Answers

The values below are approximate expected results based on the practice dataset. Use these to sanity-check your report — if your numbers are significantly different, revisit your data model or cleaning steps.

Expected data shape (after cleaning)

In-Store Transactions (clean)	~710–725 rows; date range Jan 2024 – Dec 2025; 14 unique Store IDs
Online Orders	~510 rows; 10 ship-to states; date range Jan 2024 – Dec 2025
Products	49 unique products across 7 categories
Product Reference TXT	49 products; 7 columns after splitting on | and removing header rows
Null/error rows removed	~15–25 rows across In-Store data (blank category + negative price + N/A Store ID)
Date table	730 rows (Jan 1 2024 to Dec 31 2025)

Expected visual outcomes

Line chart pattern	Moderate seasonal dip around Feb–Mar each year; stronger Q4 as year-end fitness resolutions kick in
Top revenue category	Electronics & Tech and Footwear typically lead; Nutrition & Supplements is consistent mid-tier
State distribution (in-store)	Illinois largest share (~45–50%); Ohio second (~30–35%); Indiana smallest of physical (~20–25%)
Top online ship-to states	Illinois and Michigan tend to lead online volume; Kentucky and Tennessee are smallest
Top-selling product	Wireless Earbuds Sport (ET-001) or GPS Running Watch (ET-005) typically ranks #1 in revenue
Top endorser appearing in top sellers	Tanya Osei (marathon runner) and Sierra Watts (powerlifter) products appear frequently
2024 vs 2025	2025 total sales should be moderately higher — the dataset includes natural growth built in

Expected model structure

Fact tables	2 — In-Store Transactions, Online Orders (or 1 combined if you appended)
Dimension tables	4 — DateTable, Products, StoreLocations, ProductReference (TXT)
Relationships	Date → both fact tables on Date; Products → both fact tables on Product ID; StoreLocations → StoreSales on Store ID
Key measures	Total Sales, In-Store Sales, Online Sales, Sales YTD, [Stretch] Top Product Rank
Report pages	Minimum 2; suggested: Sales Overview + Product Insights
Slicers	Minimum 1; Year slicer or State slicer recommended

Section 9 — References & Learning Resources

The resources below align with each required skill in this practice project. Reviewing them before or during the project will help you work more efficiently.

Microsoft Power BI documentation

- Power Query overview — docs.microsoft.com/power-query
- Get data from Excel — search 'Power BI get data Excel' on docs.microsoft.com
- Import delimited text files — search 'Power Query text CSV source'
- Merge queries (JOIN) in Power Query — search 'Power Query merge queries'
- Append queries (UNION) — search 'Power Query append queries'
- Create a date table in DAX — search 'Power BI CALENDAR function'
- Mark as date table — search 'Power BI mark date table'
- Model View and relationships — search 'Power BI create relationships'

DAX references

- SUM, SUMX functions — dax.guide/sum
- CALCULATE function — dax.guide/calculate
- RANKX function — dax.guide/rankx
- ALL function (used with RANKX) — dax.guide/all
- Date intelligence (TOTALYTD, SAMEPERIODLASTYEAR) — search 'DAX time intelligence'

Visualization best practices

- When to use a line chart vs. bar chart vs. pie chart — search 'Power BI choose visualization type'
- Formatting visuals in Power BI — search 'Power BI format pane'
- Slicers and cross-filtering — search 'Power BI slicer visual'
- Visual-level filters (for top N) — search 'Power BI top N filter'

A note before you begin

This practice dataset intentionally mirrors the structure and quirks of the real Capstone 3 data — two separate fact tables, a text reference file requiring a join, date formatting issues, and mixed data quality problems. If you can work through all of them here, Capstone 3 will feel familiar. Focus on understanding why each cleaning step matters, not just on getting the visual to appear. The goal is confident, explainable analysis — not just a chart on a screen.

Good luck — you've got this.